Bayesian quantile correction and synthesis for climate products

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QUESTION

Correction before synthesis

How can historical archives estimate source corrections before forecasts with different horizons are blended into one predictive distribution?

Case study: daily San Lorenzo River flow at the USGS Big Trees gauge, modeled on the $\log(1+x)$ scale with x in m^3/s. Before each origin, USGS observations and retrospective ECMWF/GloFAS and NOAA/NWS products identify source corrections. After the origin, issued ensembles and exogenous information produce source-adjusted predictive flow quantiles.

USGS observations
15-min public gauge data; daily log-flow target for state updating and held-out verification.

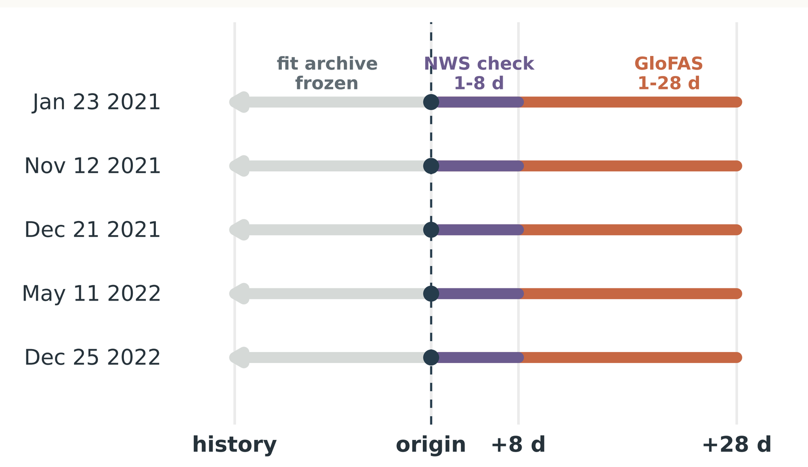
ECMWF GloFAS
retrospective alignment; daily 51-member forecast guidance to 28 days.

NOAA NWS
retrospective alignment; hourly forecasts evaluated on common daily leads 1–8.

Exogenous information
Local basin precipitation and soil moisture; the first generalized dynamic principal component (GDPC) summarizes key Pacific, Atlantic, and global climate indices.

VALIDATION

Rolling-origin holdout design



Strict temporal holdout. At each origin, fitting uses only USGS flow and retrospective products available through that date. Issued forecasts and forecast-window exogenous inputs enter after the origin; future USGS observations are reserved for scoring. Main CRPS uses GloFAS-supported days 1–28, while NWS is compared over common leads 1–8.

INFERENCE

Scalable Bayesian quantile fitting

Seven independent Dynamic Quantile Linear Model (DQLM) lanes target seven levels: 0.05, 0.20, 0.35, 0.50, 0.65, 0.80, and 0.95. The fitted quantile forecasts are checked for crossing in the application and combined into the posterior predictive distribution scored by CRPS.

7	$\leq$12,995	~2–4 h
quantile lanes	USGS archive days	fit + synthesis

Parallel quantile lanes
Each quantile level is fit as its own state-space model; the lanes can run concurrently before forecast quantiles are synthesized.

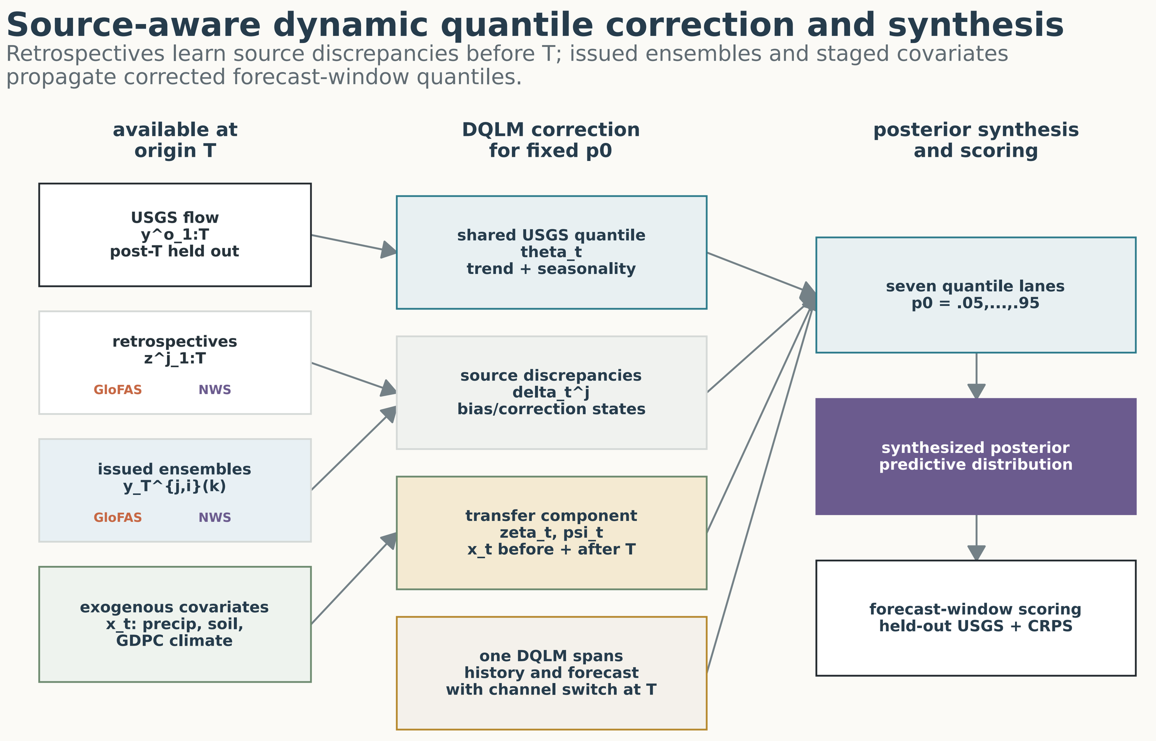
Posterior approximation
Mean-field variational Bayes (MFVB) retains closed-form Gaussian filtering and smoothing within latent-state blocks while approximating posterior dependence across blocks.

Non-conjugate updates
Variational Bayes with Laplace–Delta (VB-LD) uses local modes, curvature, and Delta-method moments for non-conjugate extended asymmetric Laplace (exAL) scale and skewness terms.

State evolution
Component discounting controls trend, multi-period seasonality, source corrections, and exogenous-covariate adjustments.

MODEL

One DQLM for source correction, then synthesis



DQLM correction stage. For a fixed quantile p_0, one Dynamic Quantile Linear Model spans the pre-origin and forecast windows: USGS observations and retrospectives identify the latent flow quantile and source-specific discrepancies before T; issued ensembles and staged covariates propagate forecast-window quantiles. Seven quantile lanes are then synthesized and scored only against held-out USGS observations.

Expanded quantile state-space model

For each target quantile p_0 and forecast origin T, the Dynamic Quantile Linear Model (DQLM) links observed flow, retrospective product histories, and issued forecast members through one latent river-flow quantile. Product-family discrepancies learn source corrections before the origin; the same source channels, with forecast-window covariates $\mathbf{x}_{T+k}$, then define corrected predictive quantiles after the origin.

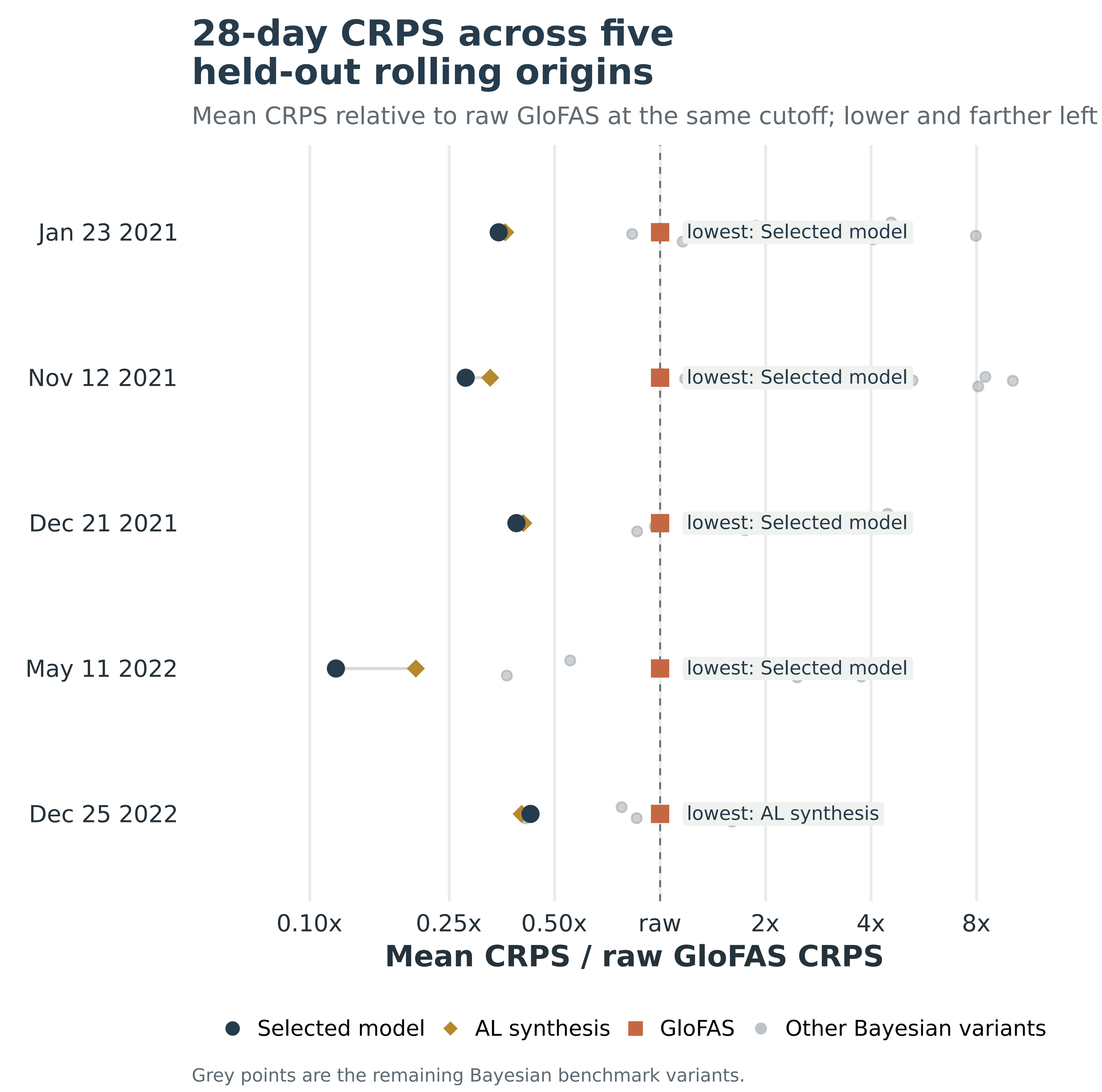
$$
\begin{array}{rlll}
\text{Observed measurements:} & y_t^o & \sim \mathrm{exAL}_{p_0}(\mathbf{F}_t'\boldsymbol{\theta}_t + \zeta_t, \sigma^o, \gamma^o)\,, & 1 \le t \le T, \\
\text{Retrospective products:} & z_t^j & \sim \mathrm{exAL}_{p_0}\left(\mathbf{F}_t'(\boldsymbol{\theta}_t + \boldsymbol{\delta}_t^j) + \zeta_t, \sigma^j, \gamma^j\right), & j \in \mathcal{J},\ 1 \le t \le T, \\
\text{Forecast products:} & y_T^{j,i}(k) & \sim \mathrm{exAL}_{p_0}\left(\mathbf{F}_{T+k}'(\boldsymbol{\theta}_{T+k} + \boldsymbol{\delta}_{T+k}^j) + \zeta_{T+k}, \sigma^j, \gamma^j\right), & j \in \mathcal{J},\ i \in \mathcal{I}_j,\ k \in \mathcal{K}_j, \\
\text{Trend and seasonal states:} & \boldsymbol{\theta}_t \mid \boldsymbol{\theta}_{t-1} & \sim \mathcal{N}\left(\mathbf{G}_t\boldsymbol{\theta}_{t-1}, \mathbf{W}_t^{\theta}\right), & 1 \le t \le T + K_{\max}, \\
\text{Source correction states:} & \boldsymbol{\delta}_t^j \mid \boldsymbol{\delta}_{t-1}^j & \sim \mathcal{N}\left(\mathbf{G}_t\boldsymbol{\delta}_{t-1}^j, \mathbf{W}_t^{\delta^j}\right), & j \in \mathcal{J},\ 1 \le t \le T + K_{\max}, \\
\text{Exogenous adjustment:} & \zeta_t \mid \zeta_{t-1}, \boldsymbol{\psi}_{t-1} & \sim \mathcal{N}\left(\lambda\zeta_{t-1} + \mathbf{x}_t'\boldsymbol{\psi}_{t-1}, w_t^{\zeta}\right), & 1 \le t \le T + K_{\max}, \\
\text{Covariate-adjustment states:} & \boldsymbol{\psi}_t \mid \boldsymbol{\psi}_{t-1} & \sim \mathcal{N}\left(\boldsymbol{\psi}_{t-1}, \mathbf{W}_t^{\psi}\right), & 1 \le t \le T + K_{\max}.
\end{array}
$$

exAL_{p_0} is the extended asymmetric Laplace working likelihood for target quantile p_0; σ and γ are source-specific scale and skewness terms. T is the forecast origin, j a product family, i an ensemble member, k a lead, and $K_{\max}$ the longest horizon. The shared quantile state is $\boldsymbol{\theta}_t$; source discrepancies/corrections are $\boldsymbol{\delta}_t^j$; and $(\zeta_t, \boldsymbol{\psi}_t)$ is the exogenous-adjustment block driven by covariates $\mathbf{x}_t$ such as precipitation, soil moisture, and the GDPC climate-index summary. Component discounting controls the evolution variances.

MAIN EVIDENCE

28-day CRPS comparison

Source-aware exAL has the lowest 28-day CRPS at four of five origins CRPS ↓



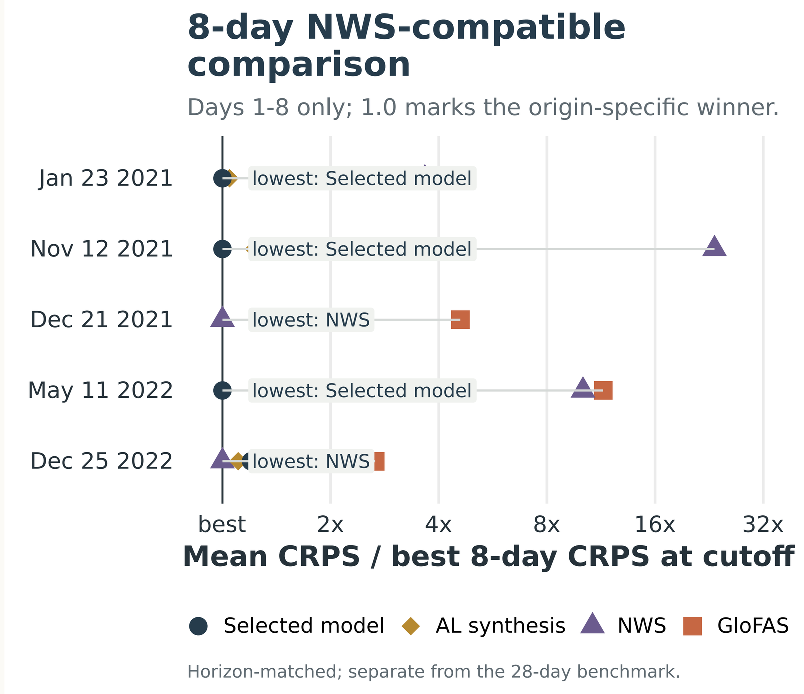
Exception: AL-M-T1 has the lowest 28-day CRPS at the 2022-12-25 origin.

How to read this panel. CRPS is the continuous ranked probability score; lower values indicate better predictive-distribution accuracy over the held-out forecast window. Ratios are relative to raw GloFAS at the same origin.

Values left of the raw GloFAS line indicate lower mean CRPS than raw GloFAS at that origin; raw NWS is evaluated only in the separate 1–8 day panel.

HORIZON

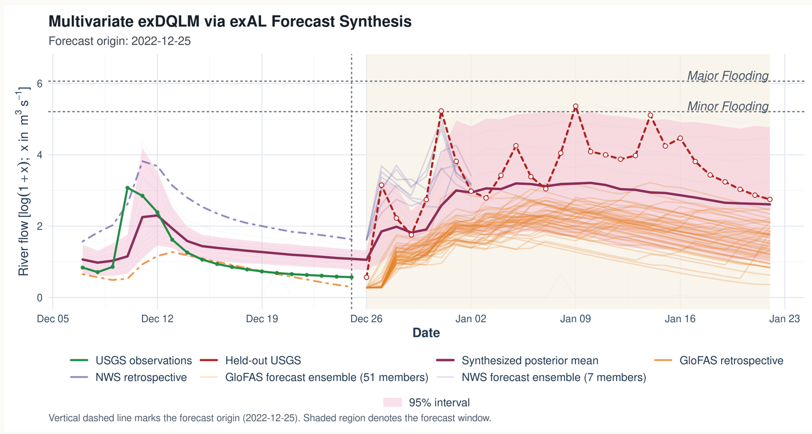
Horizon-matched NWS check



Horizon-matched evidence. Over common days 1–8, exAL-M-T1 has the lowest mean CRPS at three origins and raw NWS at two. The axis is normalized by the origin-specific best 8-day CRPS, so 1.0 marks the winner for that origin; this is separate from the 28-day benchmark.

EXAMPLE

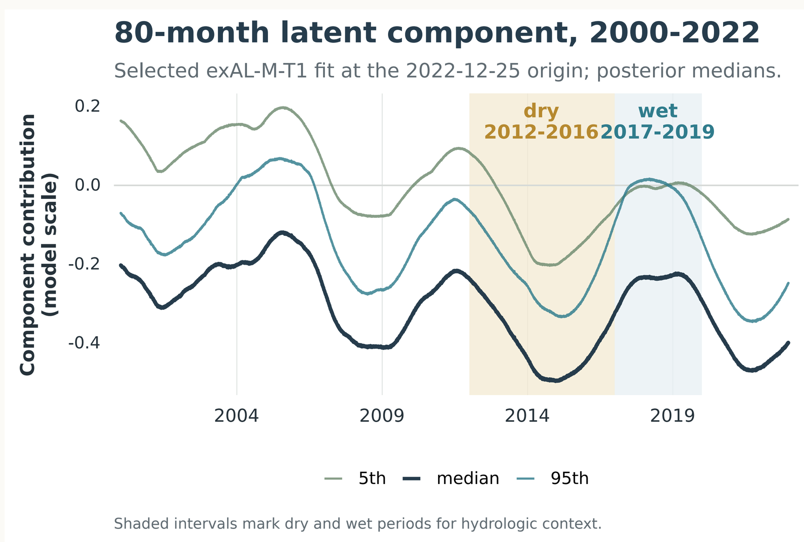
Representative forecast window



Illustrative, not aggregate validation. The selected exAL-M-T1 forecast at the 2022-12-25 origin shows how forecast inputs and posterior uncertainty evolve. At this origin AL-M-T1 has the lowest 28-day CRPS, so this panel is illustrative; aggregate evidence comes from all five origins.

DYNAMICS

80-month latent component



Retrospective diagnostic. From the selected exAL-M-T1 fit at the 2022-12-25 origin. Curves show the 5th, 50th, and 95th target-quantile components over 2000–2022; shading marks dry and wet periods for hydrologic context.

INTERPRETATION

What the evidence supports

- The source-aware exAL model (exAL-M-T1) has the lowest mean 28-day CRPS at four origins; AL-M-T1 is lowest at 2022-12-25.
- Over common days 1–8, exAL-M-T1 is lowest at three origins and raw NWS at two.
- Fitted latent states provide retrospective component diagnostics in addition to forecast-window predictive quantiles.
- The five-window design is a targeted stress test, not a continuous hindcast.

Take-home. Retrospective archives identify source-specific corrections before forecast-product blending. In this five-origin study, the source-aware exAL model leads the 28-day CRPS comparison at four origins and yields interpretable retrospective diagnostics.

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Contact: jaguir@ucsc.edu • Source logos identify data and forecast-product providers; no endorsement is implied.

Code and reproducibility
github.com/AntonioAPDL/Project1